

Hilda Adwubi Osei

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EDUCATION

Summer School in Software Development, University Service of Canada in partnership with CTVET Apr '25 — Aug '25
(GPA: Final Evaluation Pending) Accra, Ghana

- **Relevant Courses:** Data Structures and Algorithms, Object Oriented Programming, Systems Design, Version Control System, Code Implementations, Software Development Life Cycle, Software Testing, Database Management.
- Unofficially graded All A's; selected among top 9 finalists to pitch projects before national judges.

Bachelors in Industrial Engineering, Kwame Nkrumah University of Science and Technology (GPA: Jan '21 — Nov '24
Second Class Upper Division(Honors)) Kumasi, Ghana

- **Core Coursework:** Operations Research, Data Analytics, Optimization Methods, Engineering Mathematics, Production Systems, Statistics, Calculus(1,2,3)
- Integrated computer science principles(Python, algorithms, machine learning) into engineering projects to improve decision-making and efficiency.

SKILLS

Programming Languages Python, Javascript, c#, HTML/CSS

Frameworks & Libraries ReactJS, Next.js, Django, Node.js, Vite, Tailwind CSS, Express.js

AI & Data Science Tools LangChain, OpenAI GPT APIs, Llama 3.1, Gemini, Pinecone, TensorFlow(basics), Pandas, Numpy, Matplotlib

Databases & Cloud Services Firebase, Strapi, PostgreSQL, Vercel

Version Control & DevOps Git, Github, JSON, REST APIs

Research & Optimization Tools Linear Programming, Operations Research Modeling, Nonlinear Optimization

Other Competencies Technical Writing, Mentoring and Teaching Assistance, Public Speaking, International Relations, Social Development, Public Policy, Leadership

EXPERIENCE

Founder & Lead Developer

May '25 — Present

FarmDirect

Accra, Ghana

- Designed and developed **FarmDirect**, a data-driven marketplace that connects farmers directly to buyers, optimizing prices and eliminating middlemen.
- Applied **optimization modeling** and **AI** analytics to recommend fair pricing strategies and forecast market demand trends.
- Collaborated with researchers from the Ghana Atomic Energy Commission to integrate **supply chain and environmental data** for sustainable agricultural planning.
- Presented project outcomes and research findings at international forums in **Botswana, South Korea, and Belgium**.

Research & Innovation Collaborator

Jun '25 — Present

Ghana Atomic Energy Commission

Accra, Ghana

- Partnered on a **research-driven technology initiative** to improve agricultural efficiency through AI and optimization.
- Conducted system modeling and algorithm design for data collection, analysis, and visualization related to crop yield optimization.
- Supported **pilot testing and field validation** of the FarmDirect system, contributing to interdisciplinary research outputs.

Teaching and Mentorship Experience

Sep '25 — Present

National Software Development Summer School

Accra, Ghana

- Invited to support the next cohort of the **National Software Development Summer School** after ranking among top graduates.
- Delivered peer-led tutorials on **full-stack web development**, focusing on React, Firebase, and backend integration using Strapi.
- Mentored participants in project design, version control, and presentation preparation for national innovation pitches.
- Guided junior developers in best practices for **team collaboration and clean code principles**.

Software Engineering Fellow

Jul '24 — Sep '24

Headstarter AI

New York, United States (Remote)

- Built and deployed **5 AI-powered web applications** in 5 weeks using React, Next.js, Firebase, Clerk, and Vercel under agile sprints.
- Collaborated in teams of 3 to develop an **interactive customer support agent** using Next.js and a custom RAG pipeline (OpenAI + Pinecone).
- Co-developed a **SaaS product for AI flashcards** using the Llama 3.1 model via Groq API, integrating Stripe for custom pricing and payments.

- Engaged in weekly mentorship sessions with engineers from **Google, Y Combinator, and Stanford**, applying CI/CD and product scaling principles.

PROJECTS

AI Resume Builder (Team of 3) [Link](#)

- *React, Vite, Tailwind CSS, Strapi, Clerk*
 - Developed an AI-powered platform that generates professional resumes with dynamic content suggestions and real-time customization.
 - Implemented secure authentication using **Clerk**, backend content management with **Strapi**, and responsive frontend interface using **React (Vite)** and **Tailwind CSS**.

AI Rate Professor (Team of 3) [Link](#)

- *LangChain, Pinecone, OpenAI GPT, JavaScript*
 - Engineered a **Retrieval-Augmented Generation (RAG) pipeline** integrating **Pinecone vector database** and **LangChain** for contextual professor ratings and analysis.
 - Leveraged **OpenAI GPT models** for semantic understanding and automated feedback synthesis based on real-time academic data.

Customer Support Chatbot (Team of 3) [Link](#)

- *Next.js, Gemini, Pinecone*
 - Built a customer service chatbot leveraging **Gemini LLM** and **Pinecone vector search** for intelligent, context-aware query handling.
 - Deployed a **Next.js** interface enabling interactive, multilingual, and dynamic conversation management for users.

LEADERSHIP & GLOBAL ENGAGEMENT

Delegate, Innovation Fair for Africa-Europe Collaboration(Belgium 2025)

Invited to present the **FarmDirect** initiative at the Innovation Fair promoting Africa–Europe technology partnerships and sustainable development.

Networked with EU and AU innovation leaders to explore potential research and funding collaborations for agricultural technology.

Delegate, International "We, The Youth" Forum (Botswana, South Korea, 2025)

One of **150 explorers selected(out of 27,00+ applicants)** for this fully funded forum focused on youth-led innovation, governance, and sustainable development.

Presented **FarmDirect**, an AI-powered agricultural platform, during the regional conference in Botswana and later at the **Global Youth Summit in South Korea**, representing Ghana in global policy discussions.

Participant, Human Rights Advocate Fellowship (Budapest, Hungary, 2025)

Selected for a **fully funded human rights advocacy fellowship** in Hungary focusing on digital rights, policy advocacy, and technology’s role in promoting social justice.

Engaged in cross-cultural dialogue on freedom of expression and technology’s intersection with global human rights issues.

Leadership Diary Counselor, Youth Arise Organization (2020 - Present)

Selected as part of the 2020 cohort of the Leadership Diary Program and later appointed as a **Counselor** to mentor new participants on leadership, innovation, and personal development.

Facilitated workshops on goal setting, emotional intelligence, and community impact for over **100 youth participants** across Ghana.

AWARDS

Academic Distinction - Industrial Engineering (Graduated 2024)

Kwame Nkrumah University of Science and Technology

Graduated with **Second Class Upper Division Honors**, consistently ranking among the top students in the department.

Innovation Grant Winner (2025)

National Software Development Summer School

Awarded **€10,000** to scale the *FarmDirect* project after emerging as one of **9 finalists** selected to pitch before a national judging panel.

Research Presentation

International "We, The Youth" Forum (Botswana & South Korea)

Invited to present *FarmDirect* and research on **AI-driven food system optimization** at both the regional and global youth summits.

Youth Leadership Recognition

Youth Arise Organization (2020 - Present)

Honored for sustained service as a **Leadership Diary Counselor**, mentoring cohorts of young leaders and supporting national youth empowerment programs.